INCH-POUND
MIL-DTL-83507/17A
25 July 2025
SUPERSEDING
MIL-T-83507/17(AS)
7 July 1978

DETAIL SPECIFICATION

Tool Kits, Electrical Connectors, Contacts, and Connector and Cable Accessories, Tool Kit for SAE-AS22520/2 Crimping Tool and Positioners

Inactive for new design as of 01 Sep 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the tool kit described herein shall consist of this document and the latest issue of Specification MIL-DTL-83507.

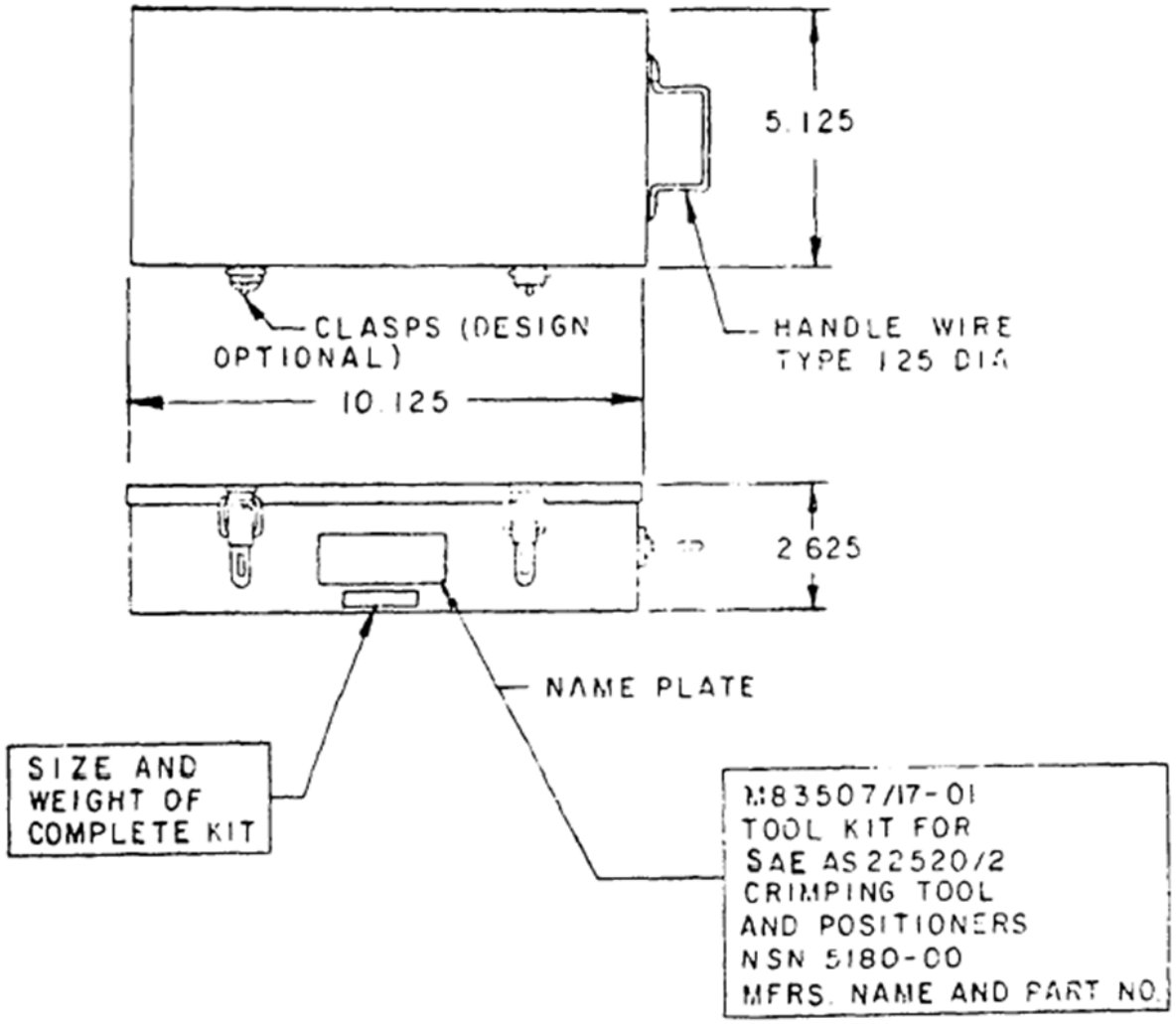


FIGURE 1 Metal case. Tool kit for SAE-AS22520/2 crimping tool and positioners.

AMSC N/A FSC 5180

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

NOTES:

1. Part number M83507/17-01 includes case and tools.
2. Part number M83507/17-02 for case only (tools not included).

Parts list

(1)	M83507/17-2	Tool case assembly.
(1)	M22520/2-01	Crimp tool frame.
(1)	M22520/2-02	Positioner.
(1)	M22520/2-03	Positioner.
(1)	M22520/2-04	Positioner.
(1)	M22520/2-05	Positioner.
(1)	M22520/2-06	Positioner.
(1)	M22520/2-07	Positioner.
(1)	M22520/2-08	Positioner.
(1)	M22520/2-09	Positioner.
(1)	M22520/2-10	Positioner.
(1)	M22520/3-1	In-service inspection gage.
(1 set)		Contents charts, instruction sheets, and tool selection chart

FIGURE 1 Metal case. Tool kit for SAE-AS22520/2 crimping tool and positioners. – Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

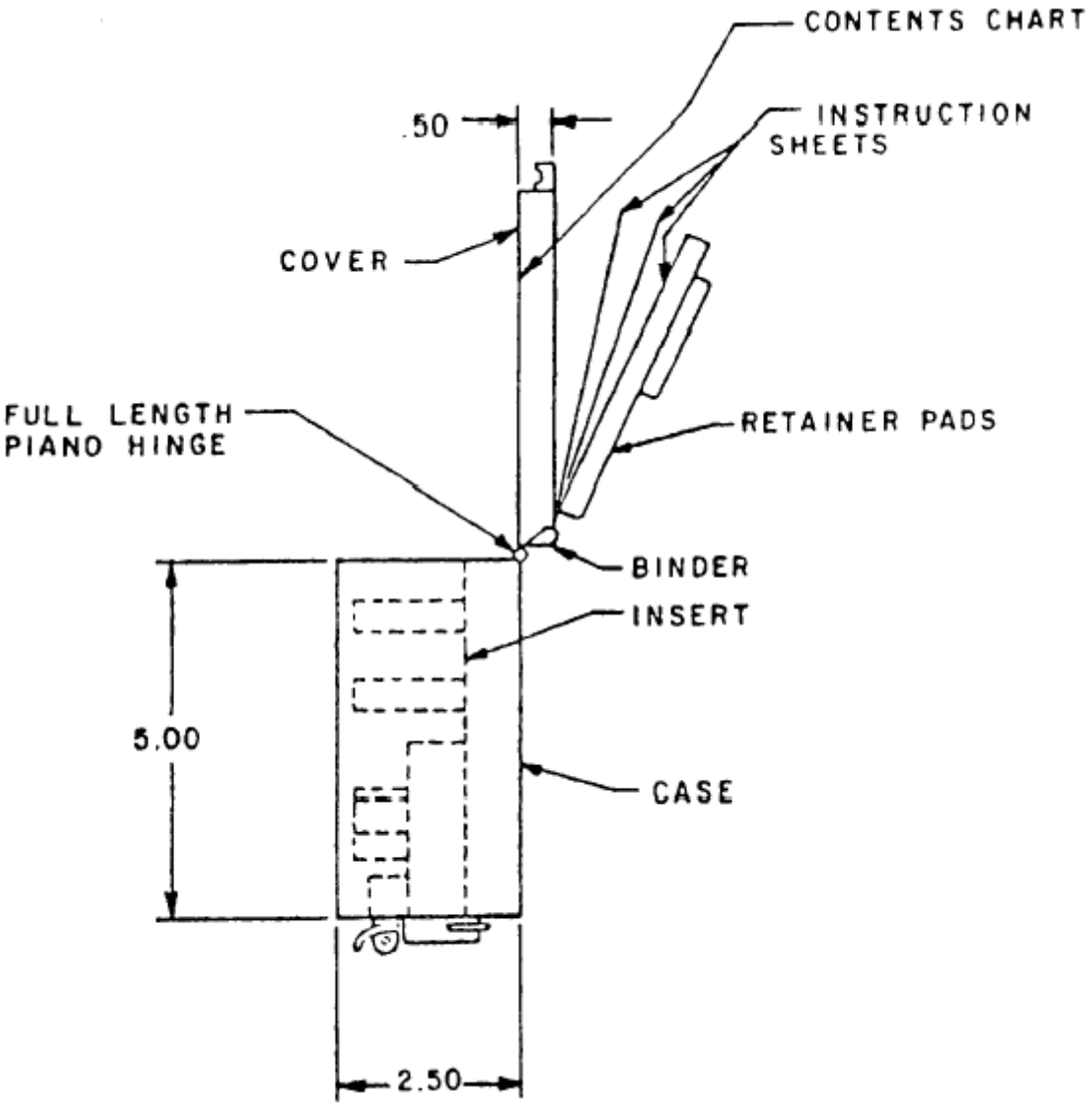


FIGURE 2 Metal case. Tool case assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

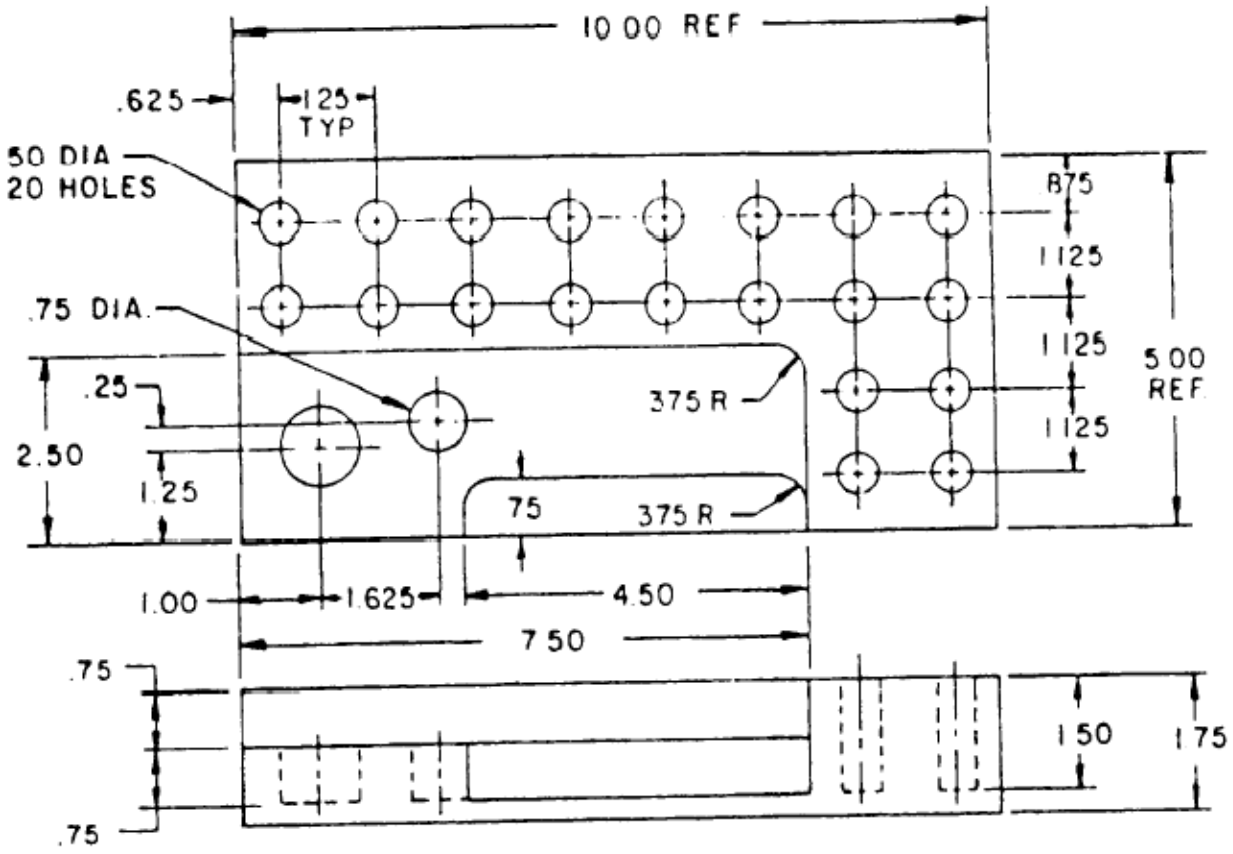


FIGURE 3 Metal case. Insert.

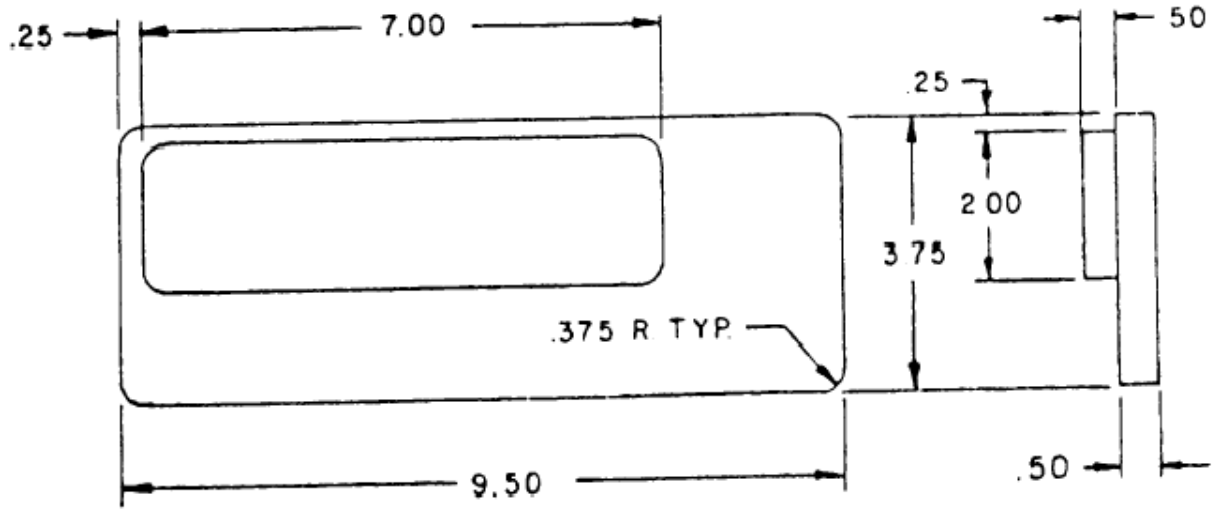


FIGURE 4 Metal case. Retainer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

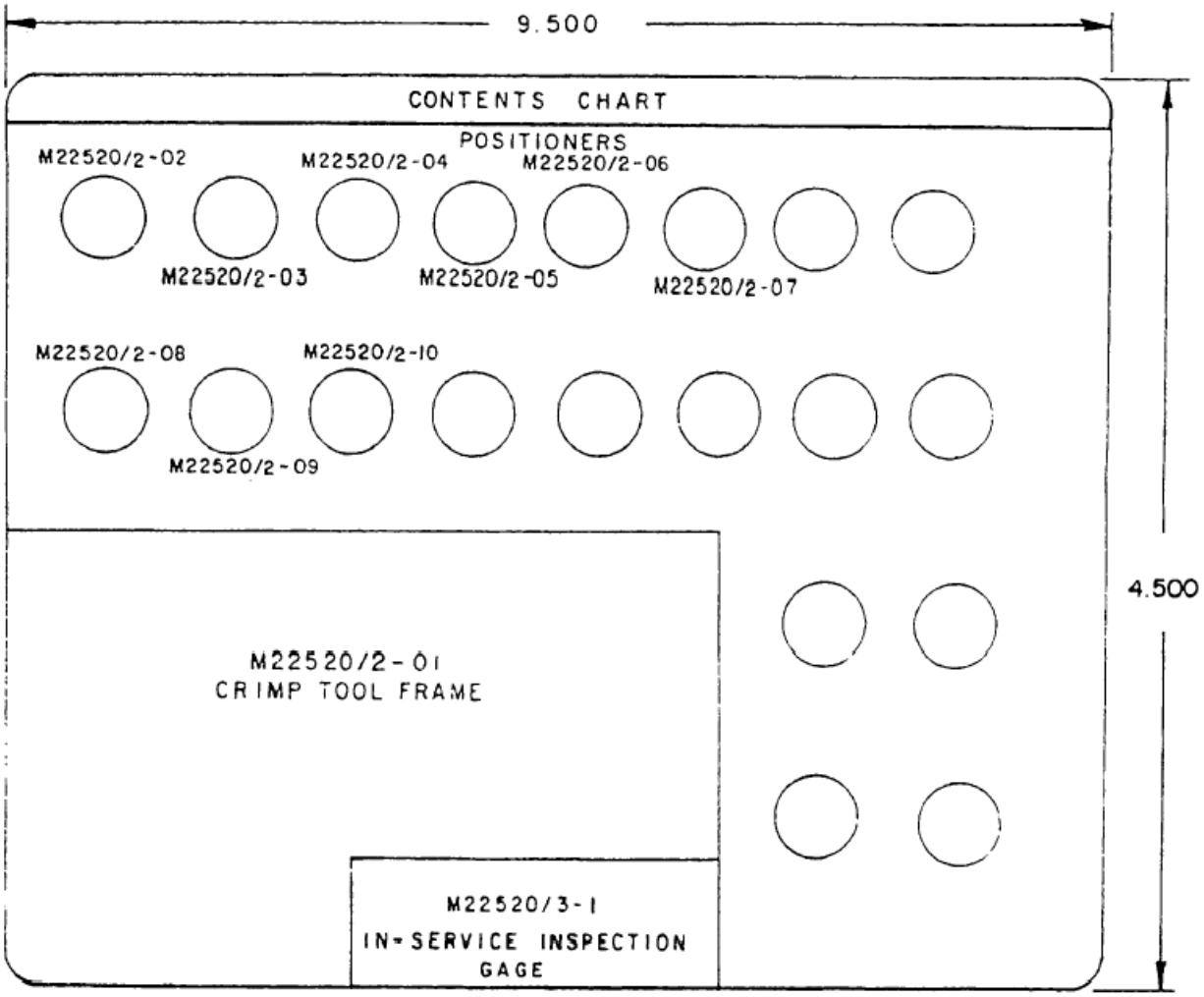


FIGURE 5 Metal case. Contents chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

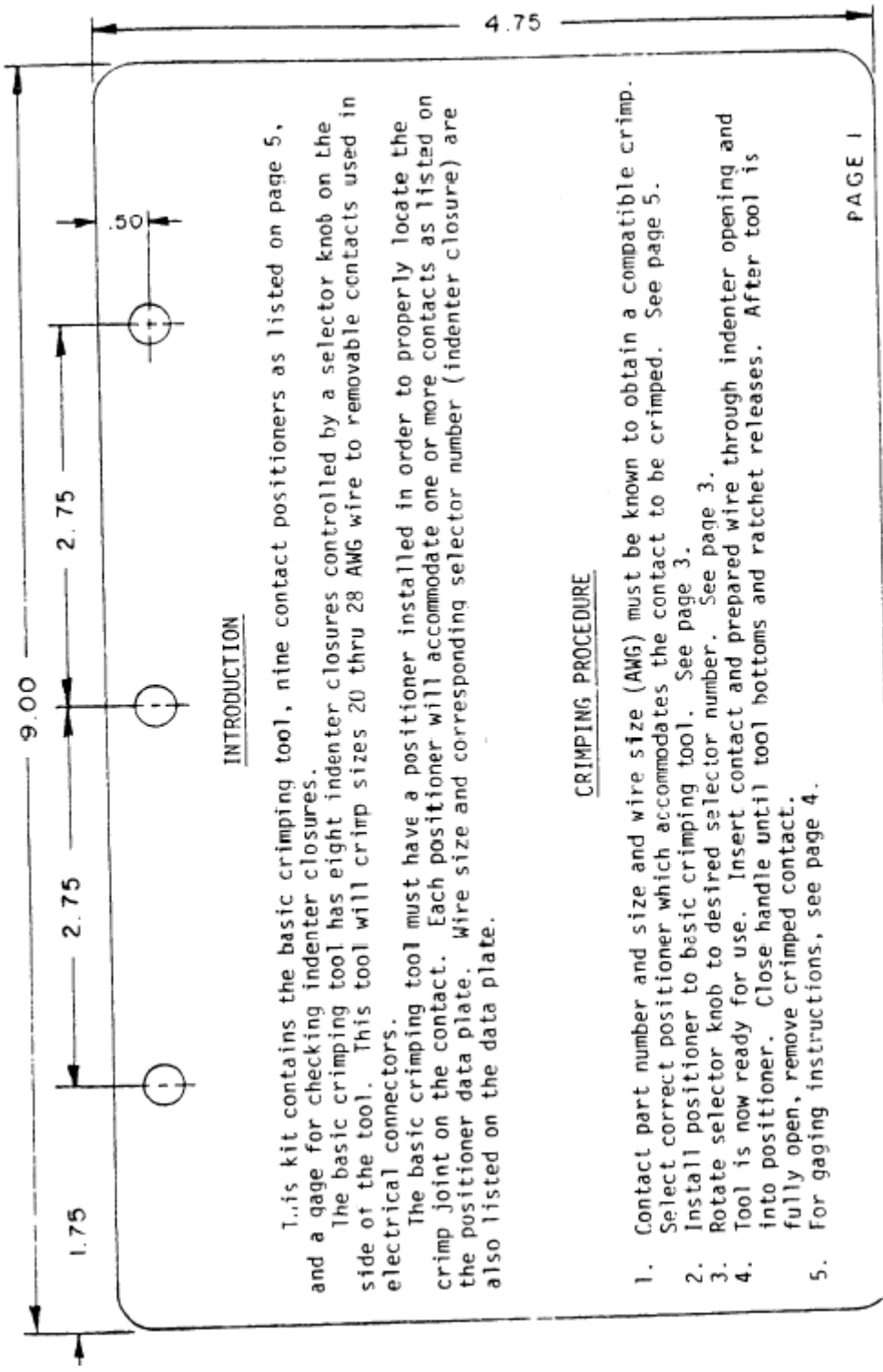


FIGURE 6 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

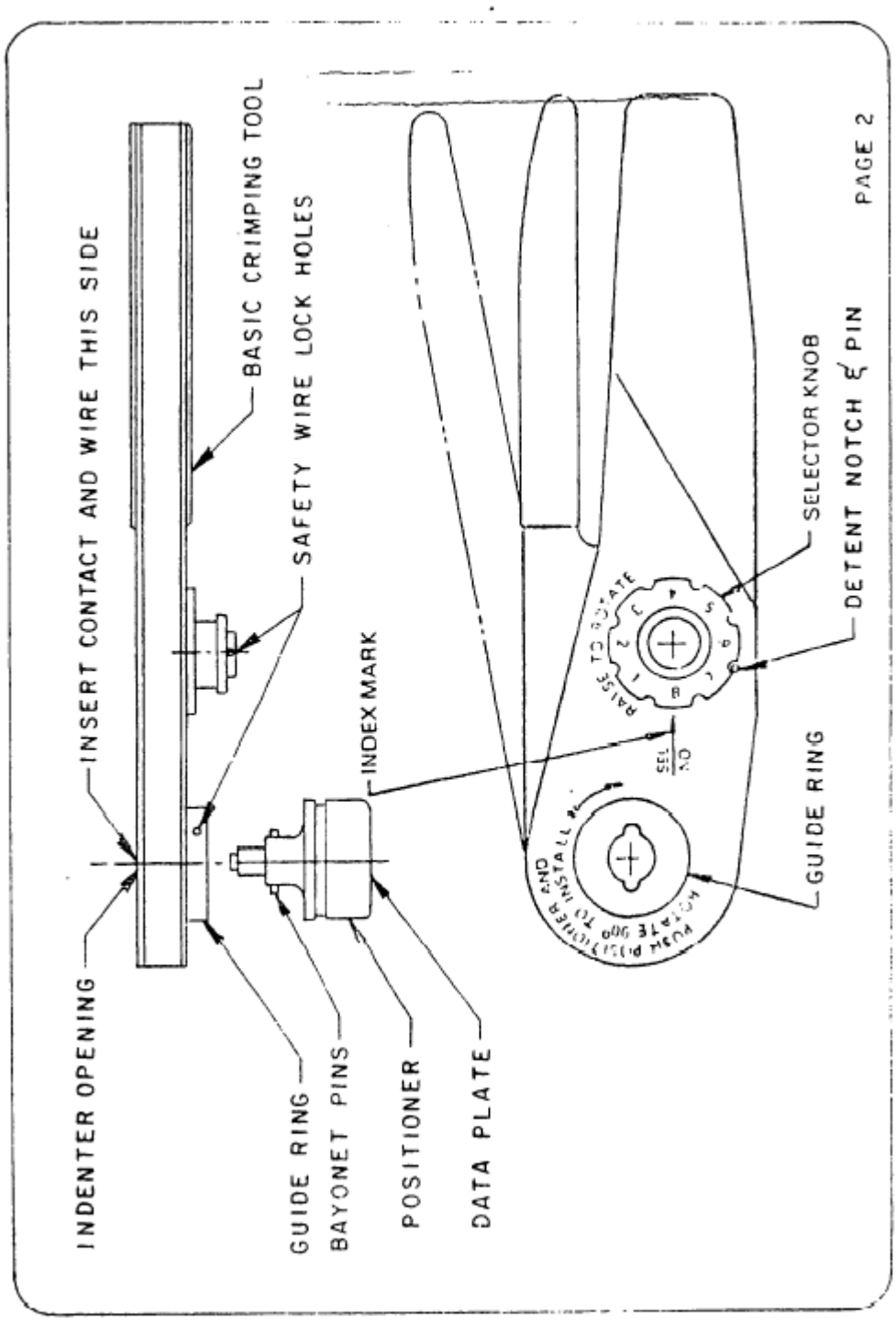
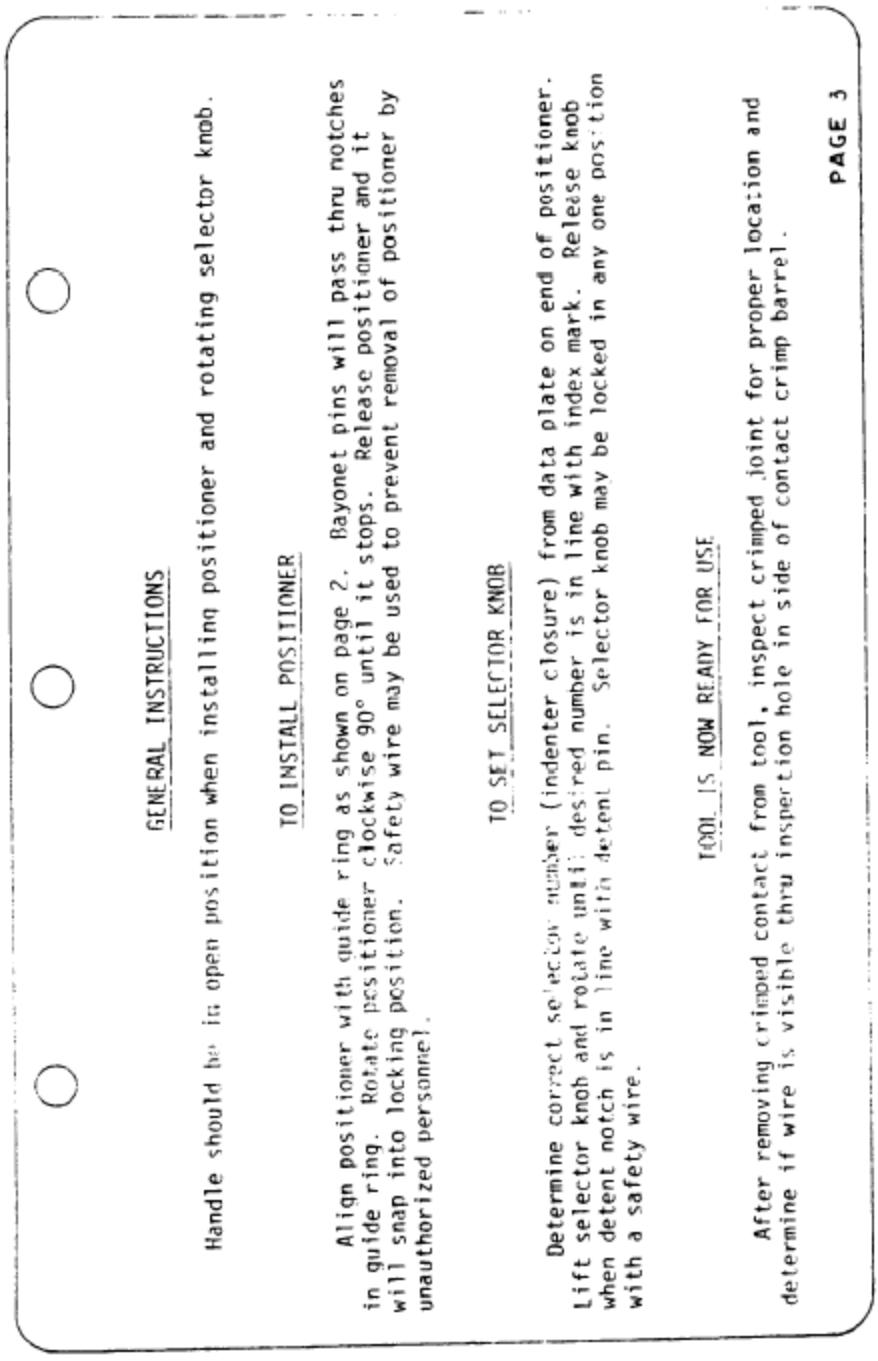


FIGURE 7 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.



GENERAL INSTRUCTIONS

Handle should be in open position when installing positioner and rotating selector knob.

TO INSTALL POSITIONER

Align positioner with guide ring as shown on page 2. Bayonet pins will pass thru notches in guide ring. Rotate positioner clockwise 90° until it stops. Release positioner and it will snap into locking position. Safety wire may be used to prevent removal of positioner by unauthorized personnel.

TO SET SELECTOR KNOB

Determine correct selection number (indenter closure) from data plate on end of positioner. Lift selector knob and rotate until desired number is in line with index mark. Release knob when detent notch is in line with detent pin. Selector knob may be locked in any one position with a safety wire.

TOOL IS NOW READY FOR USE

After removing crimped contact from tool, inspect crimped joint for proper location and determine if wire is visible thru inspection hole in side of contact crimp barrel.

PAGE 3

FIGURE 8 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

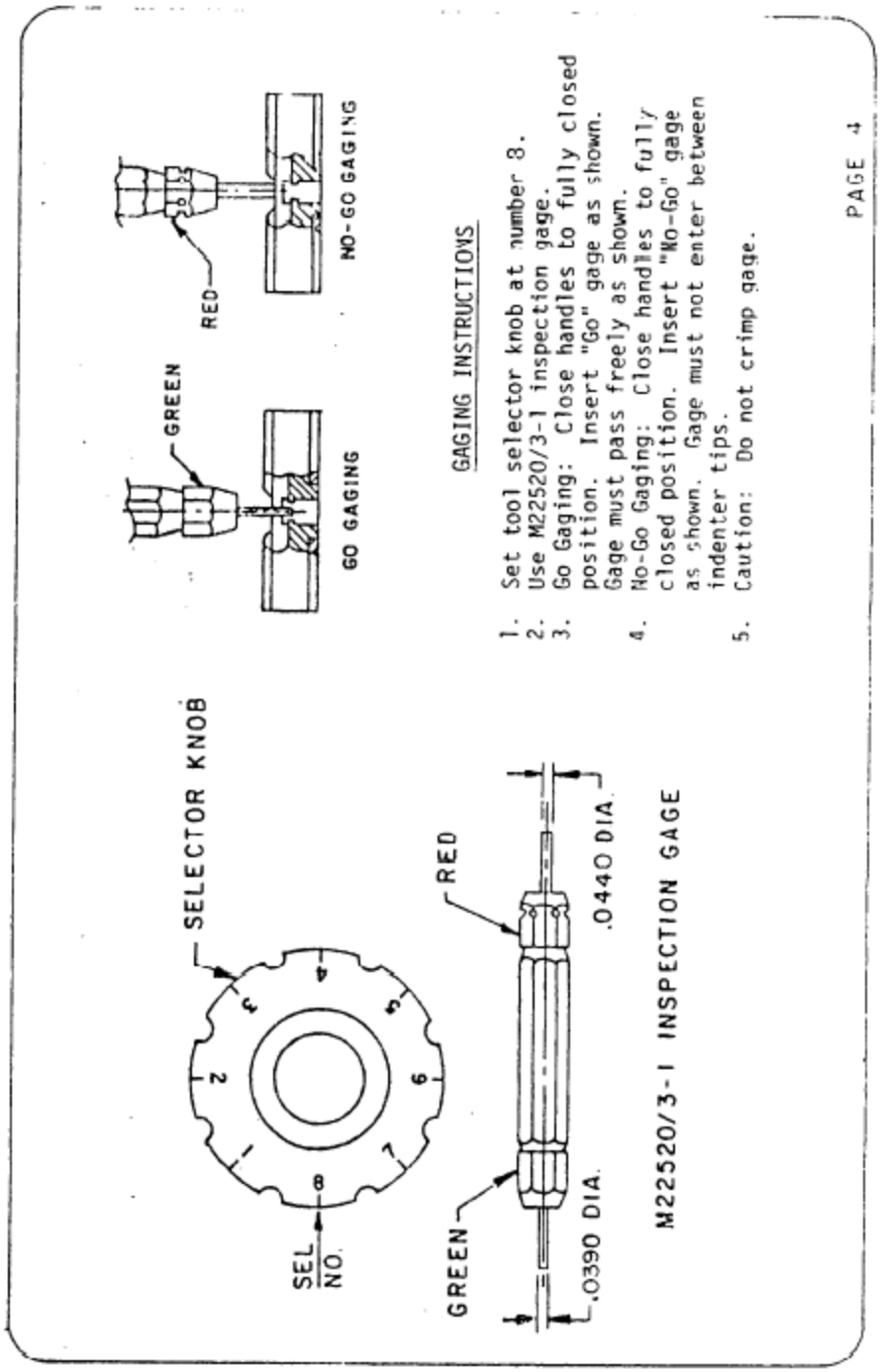


FIGURE 9 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

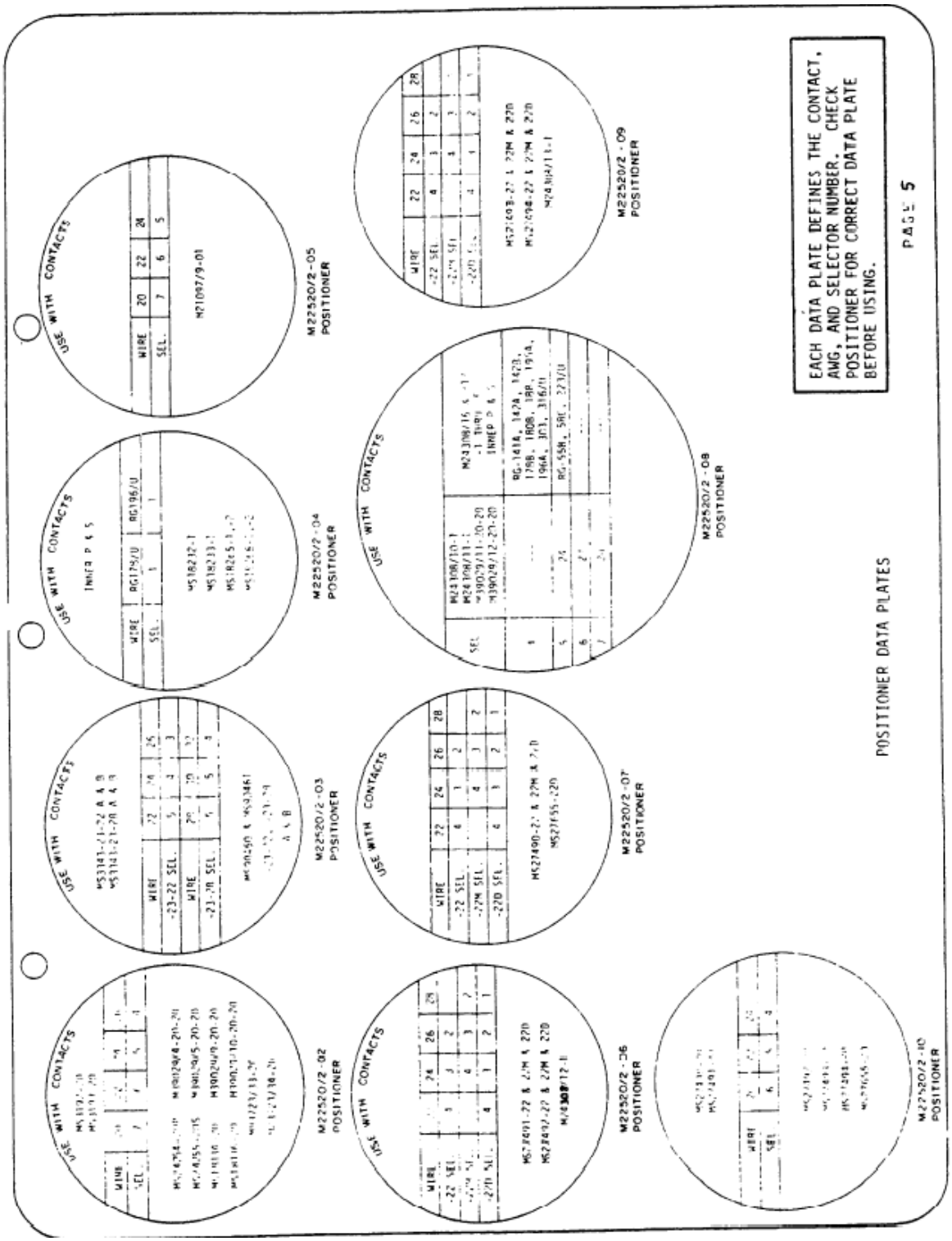


FIGURE 10 Metal case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

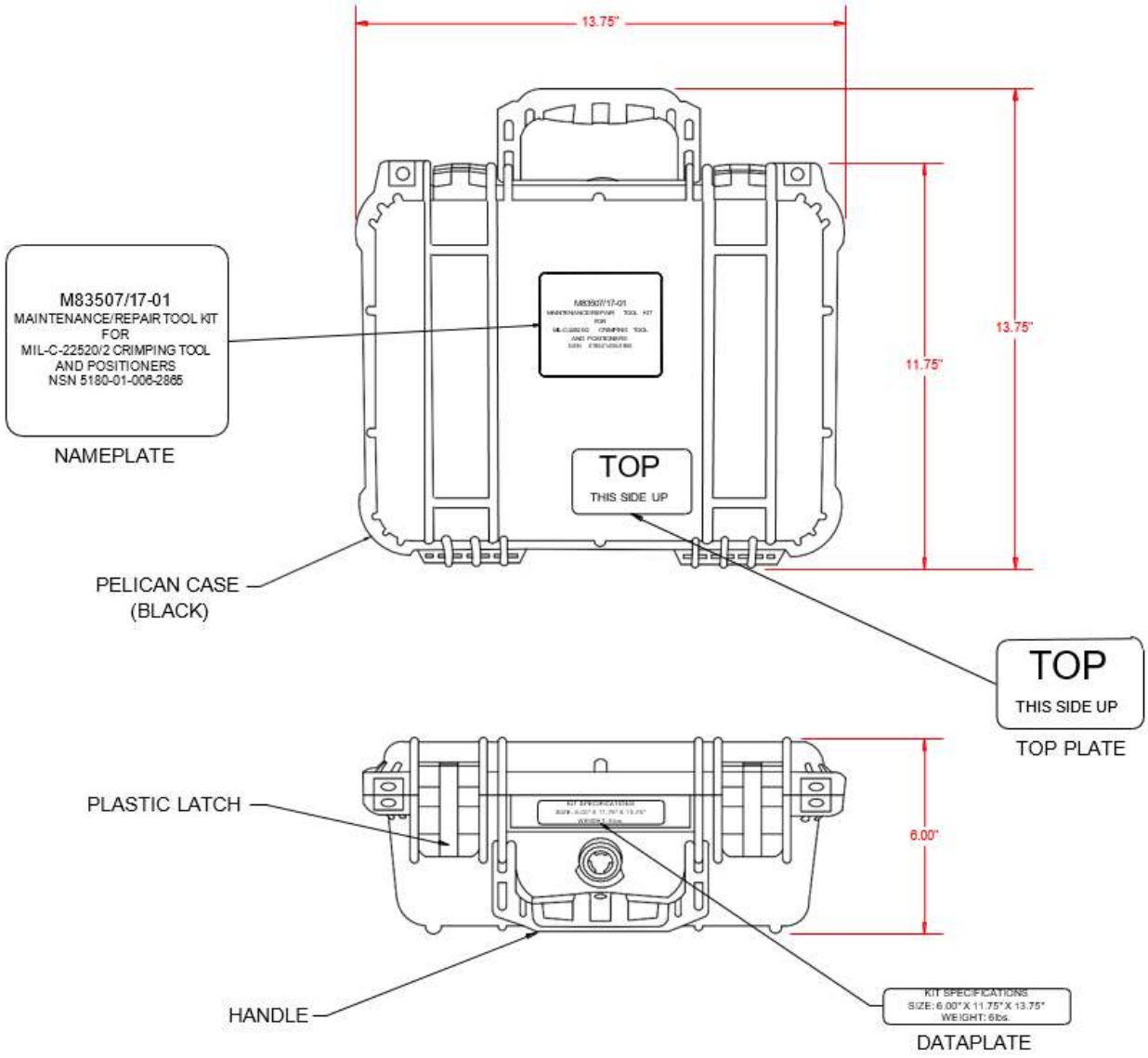


FIGURE 11 Plastic case. TOOL KIT FOR GENERAL PURPOSE BACKSHELL HARDWARE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

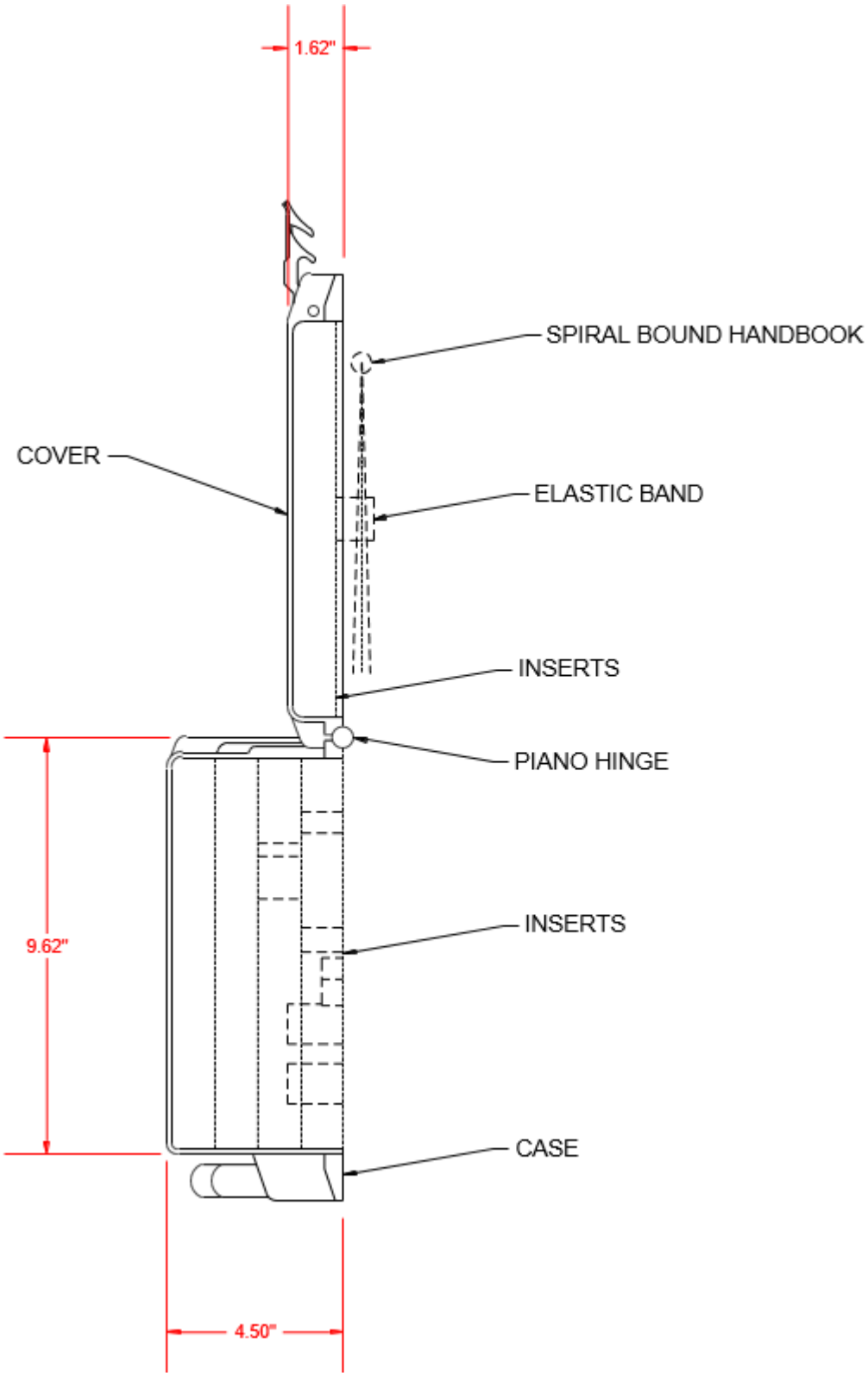


FIGURE 12 Plastic case. TOOL CASE ASSEMBLY.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

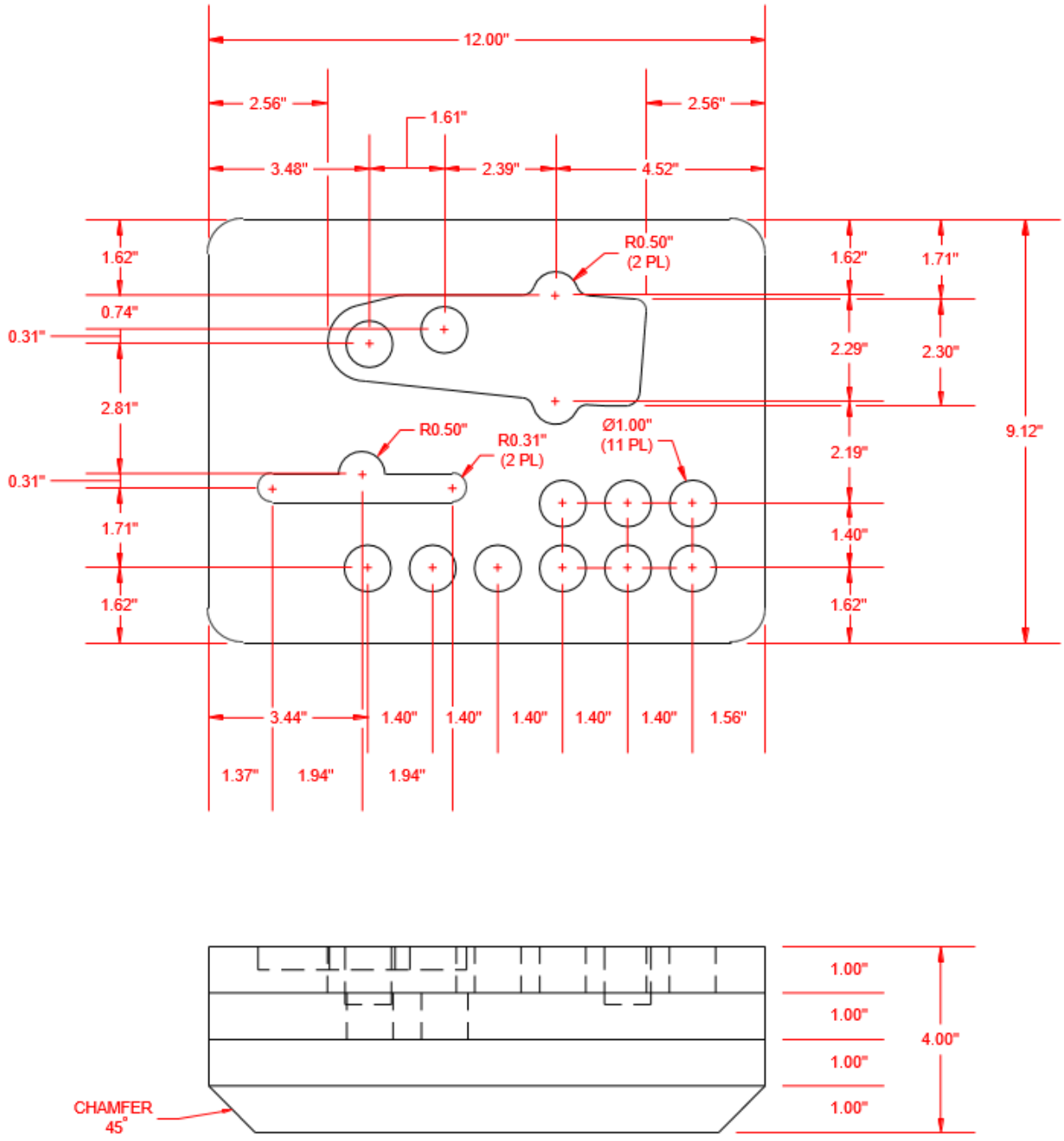


FIGURE 13 Plastic case. INSERT, BOTTOM OF CASE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

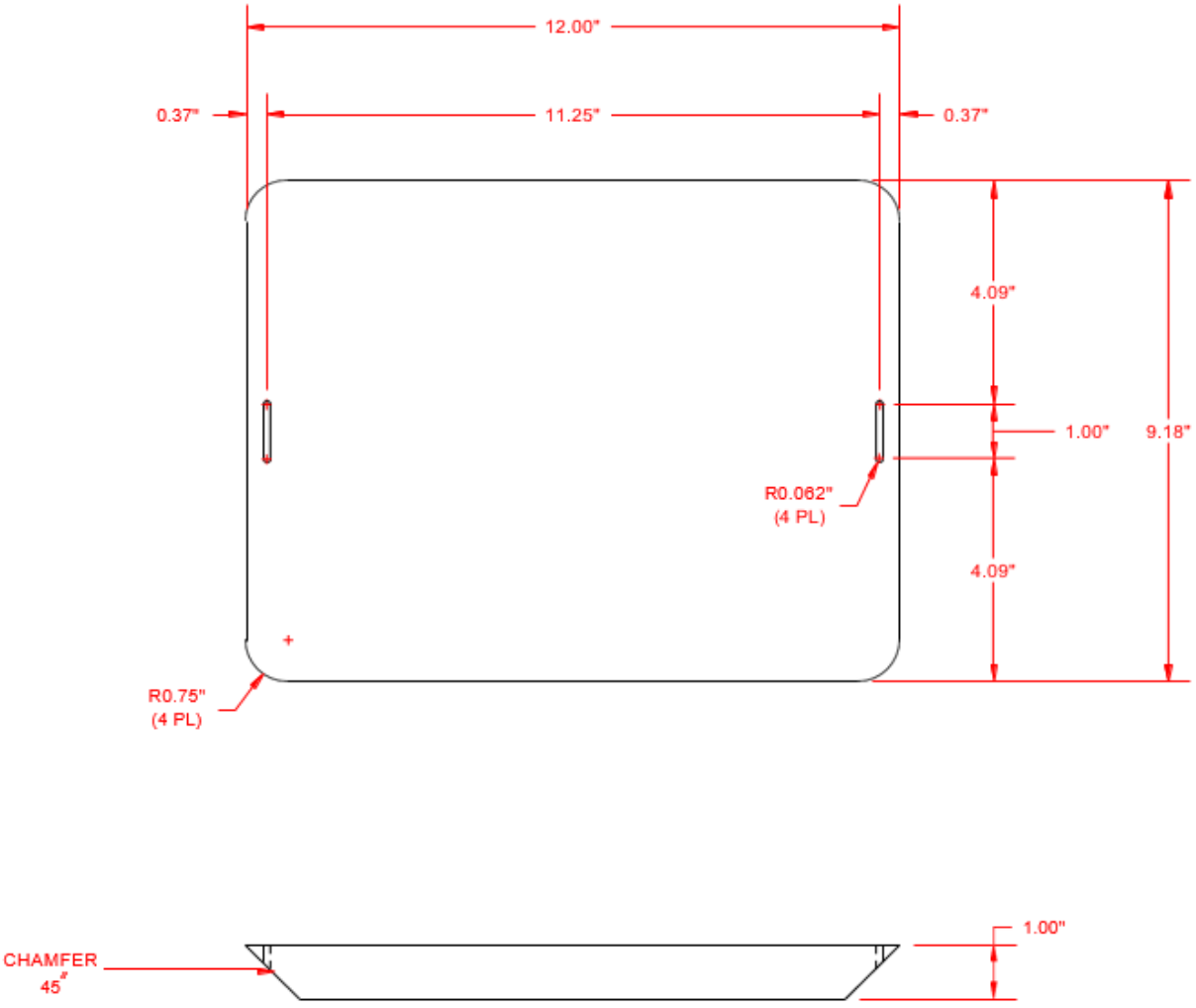


FIGURE 14 Plastic case. INSERT, TOP OF CASE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

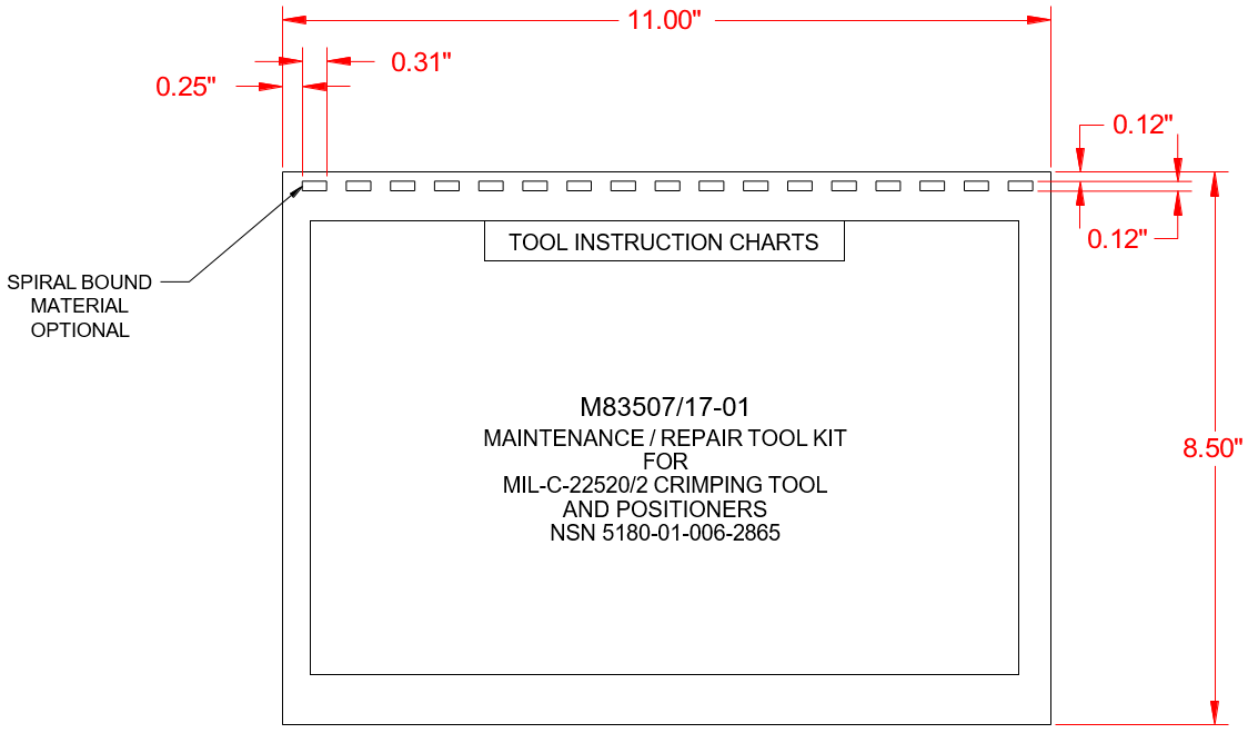


FIGURE 15 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

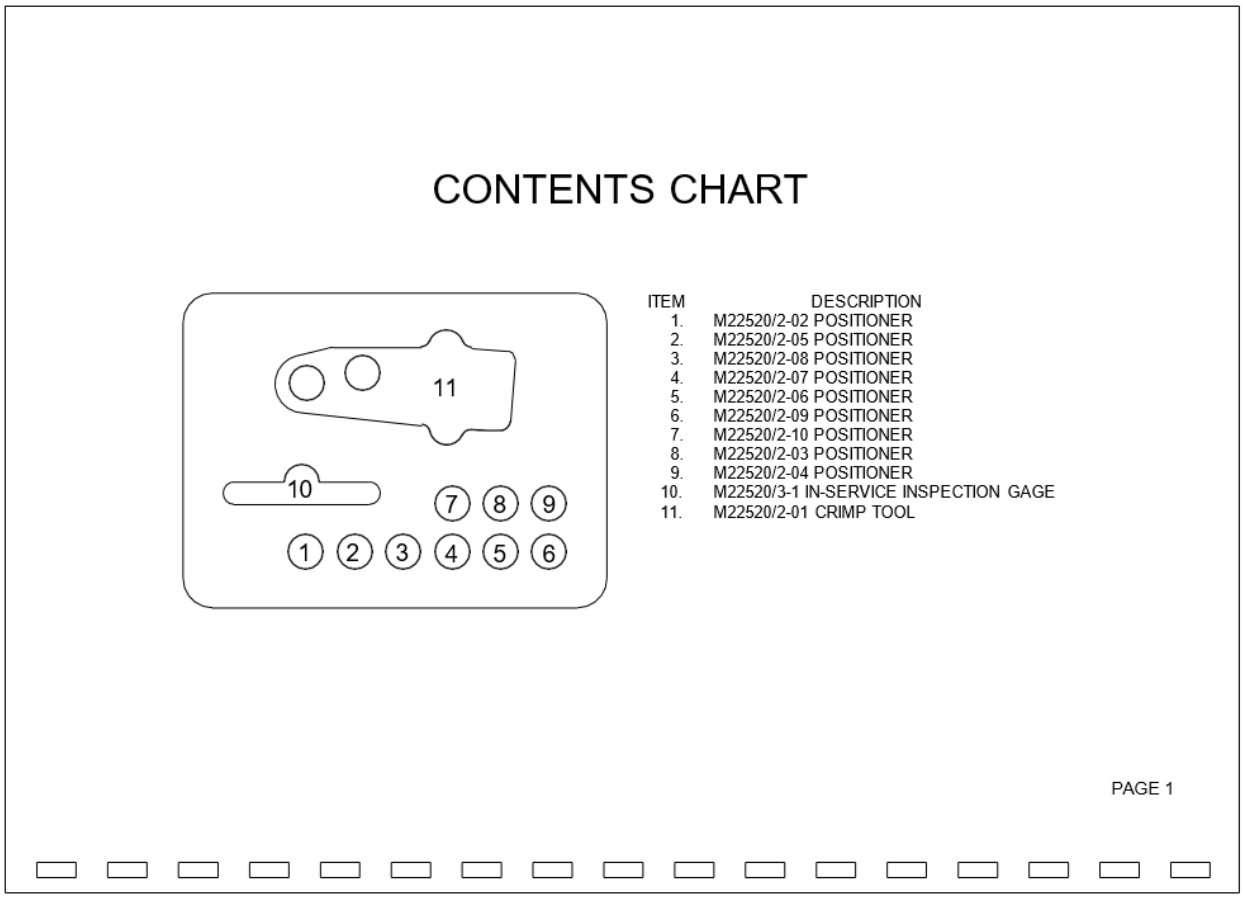
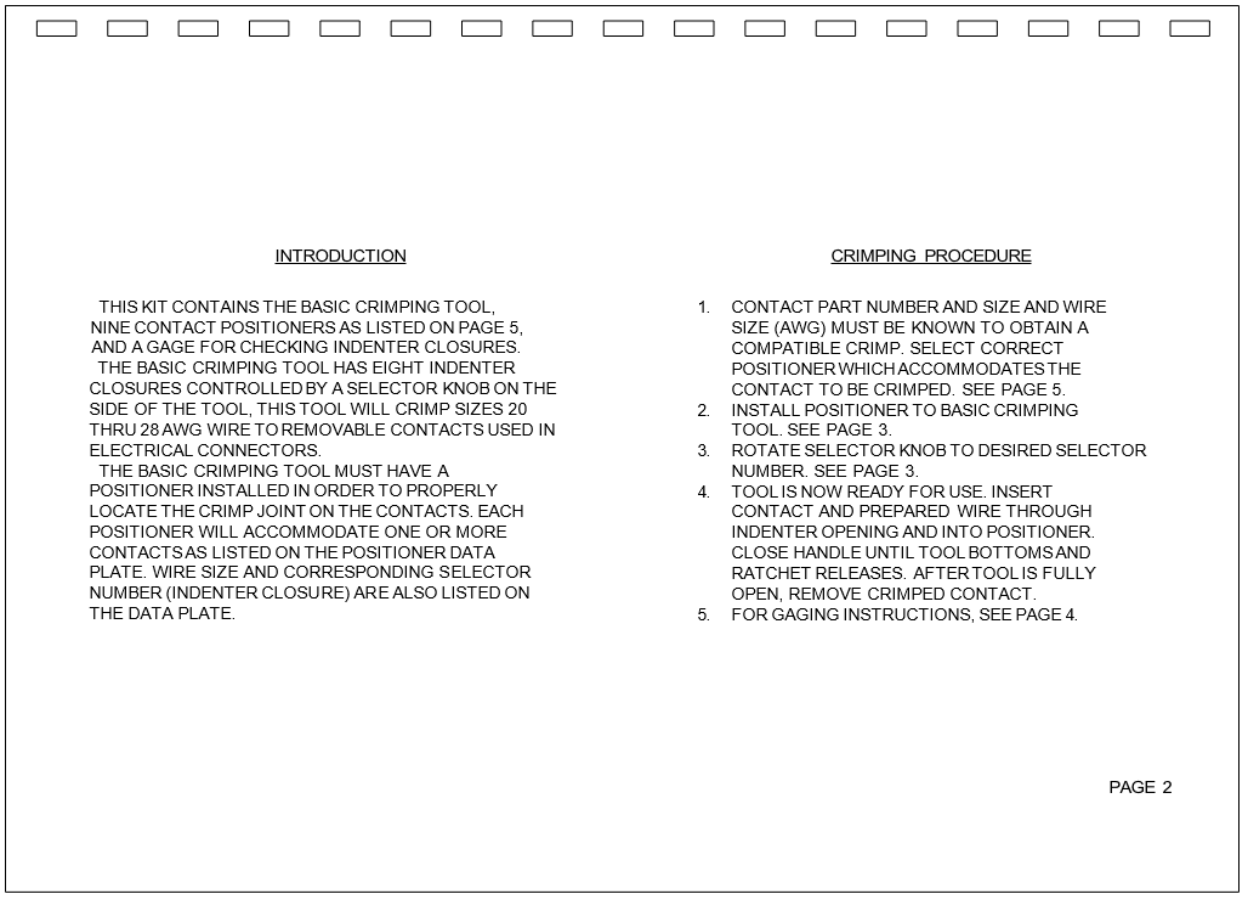


FIGURE 16 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.



INTRODUCTION

THIS KIT CONTAINS THE BASIC CRIMPING TOOL, NINE CONTACT POSITIONERS AS LISTED ON PAGE 5, AND A GAGE FOR CHECKING INDENTER CLOSURES.

THE BASIC CRIMPING TOOL HAS EIGHT INDENTER CLOSURES CONTROLLED BY A SELECTOR KNOB ON THE SIDE OF THE TOOL, THIS TOOL WILL CRIMP SIZES 20 THRU 28 AWG WIRE TO REMOVABLE CONTACTS USED IN ELECTRICAL CONNECTORS.

THE BASIC CRIMPING TOOL MUST HAVE A POSITIONER INSTALLED IN ORDER TO PROPERLY LOCATE THE CRIMP JOINT ON THE CONTACTS. EACH POSITIONER WILL ACCOMMODATE ONE OR MORE CONTACTS AS LISTED ON THE POSITIONER DATA PLATE. WIRE SIZE AND CORRESPONDING SELECTOR NUMBER (INDENTER CLOSURE) ARE ALSO LISTED ON THE DATA PLATE.

CRIMPING PROCEDURE

1. CONTACT PART NUMBER AND SIZE AND WIRE SIZE (AWG) MUST BE KNOWN TO OBTAIN A COMPATIBLE CRIMP. SELECT CORRECT POSITIONER WHICH ACCOMMODATES THE CONTACT TO BE CRIMPED. SEE PAGE 5.
2. INSTALL POSITIONER TO BASIC CRIMPING TOOL. SEE PAGE 3.
3. ROTATE SELECTOR KNOB TO DESIRED SELECTOR NUMBER. SEE PAGE 3.
4. TOOL IS NOW READY FOR USE. INSERT CONTACT AND PREPARED WIRE THROUGH INDENTER OPENING AND INTO POSITIONER. CLOSE HANDLE UNTIL TOOL BOTTOMS AND RATCHET RELEASES. AFTER TOOL IS FULLY OPEN, REMOVE CRIMPED CONTACT.
5. FOR GAGING INSTRUCTIONS, SEE PAGE 4.

PAGE 2

FIGURE 17 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

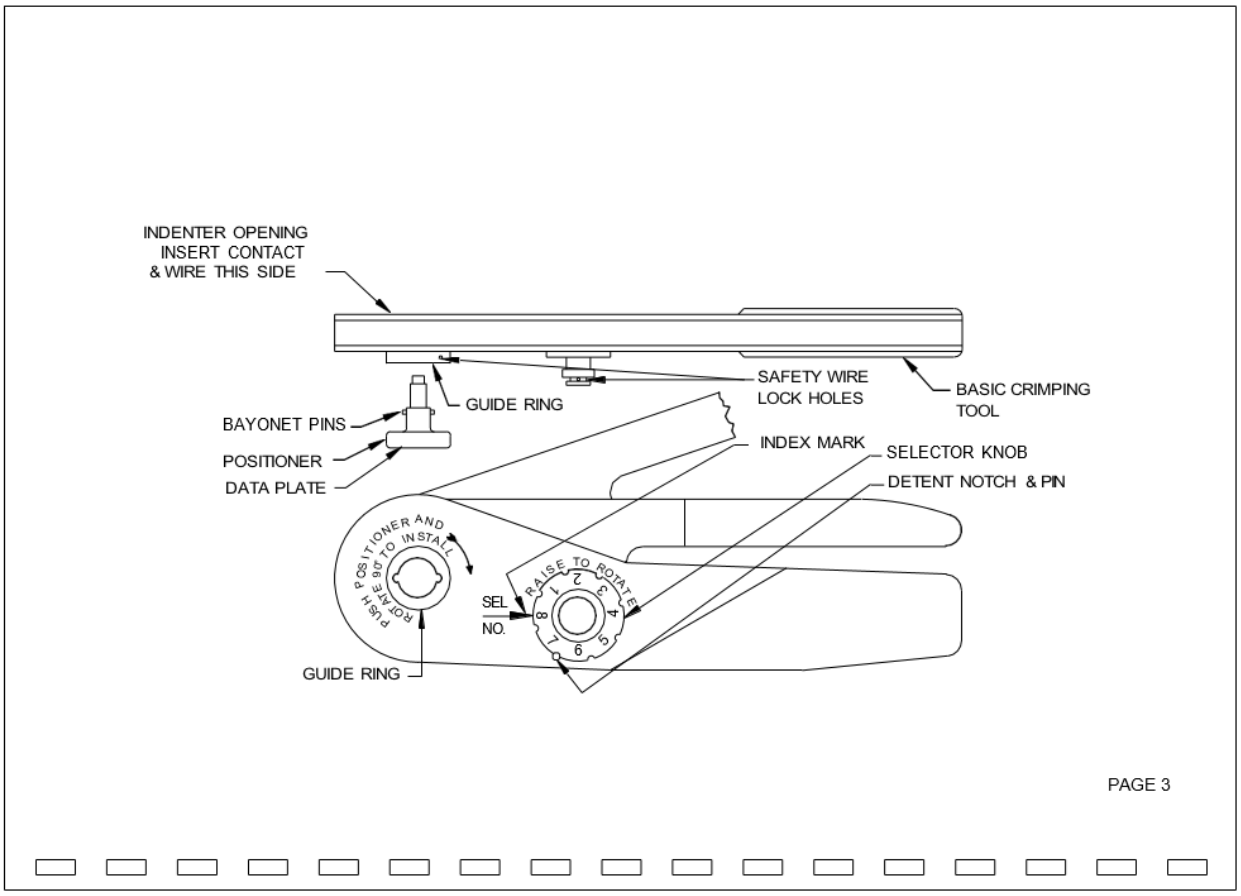


FIGURE 18 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

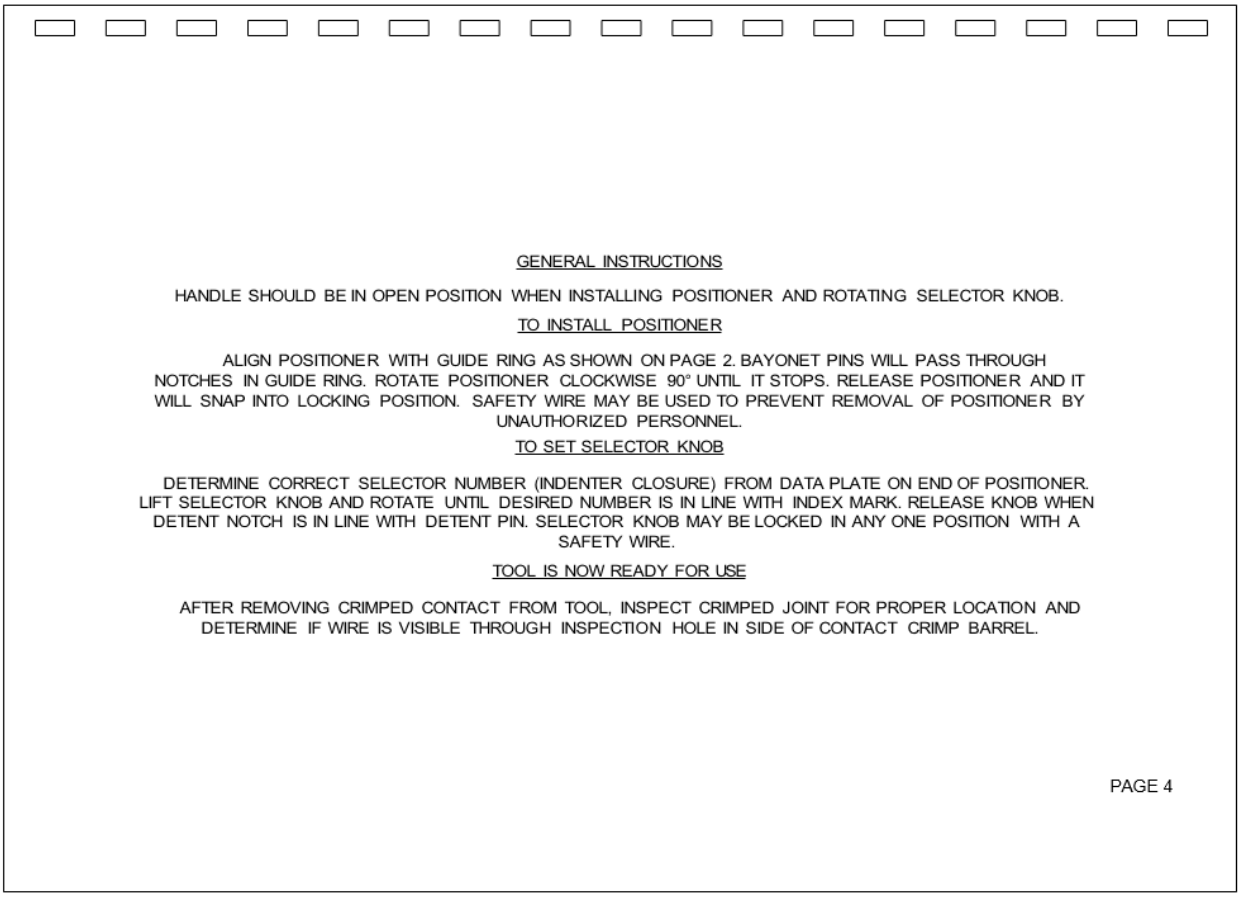


FIGURE 19 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

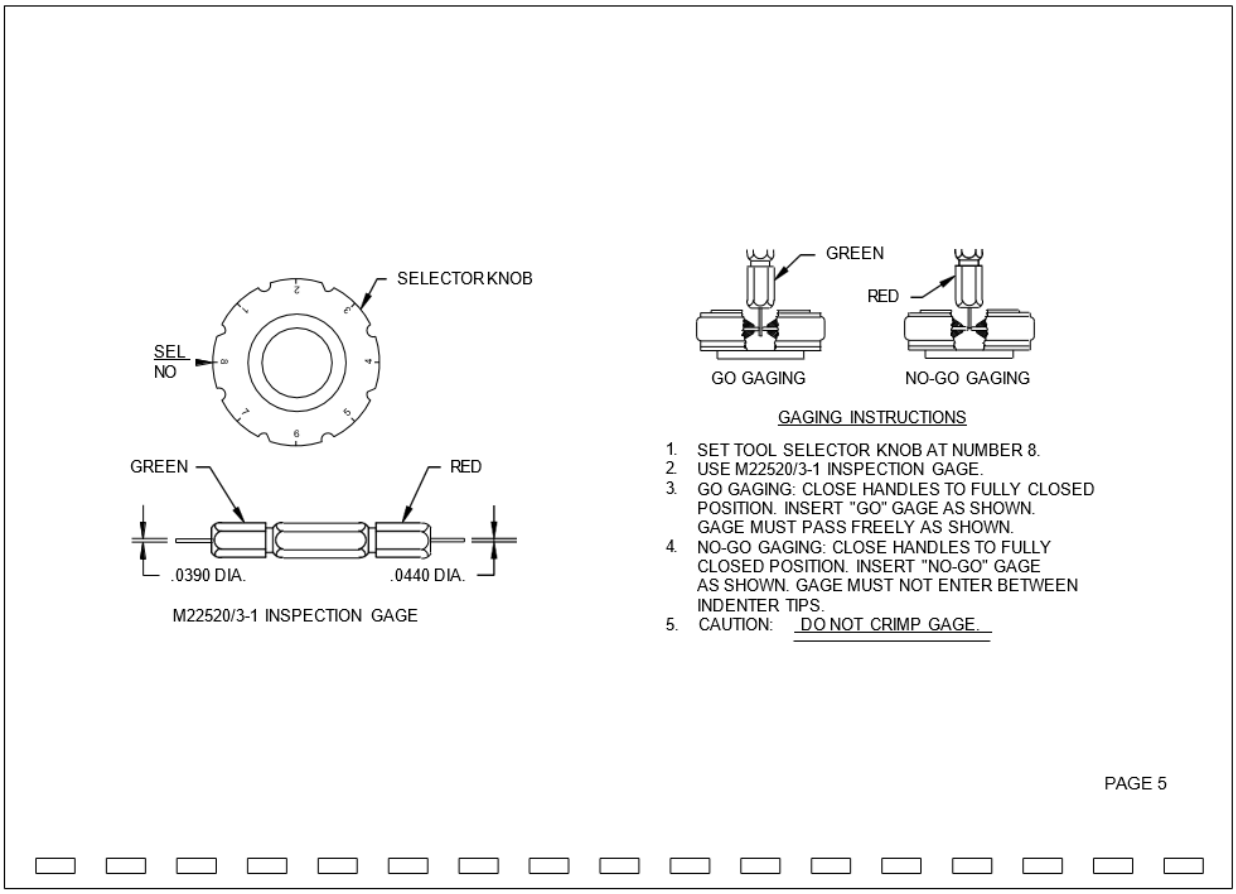


FIGURE 20 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

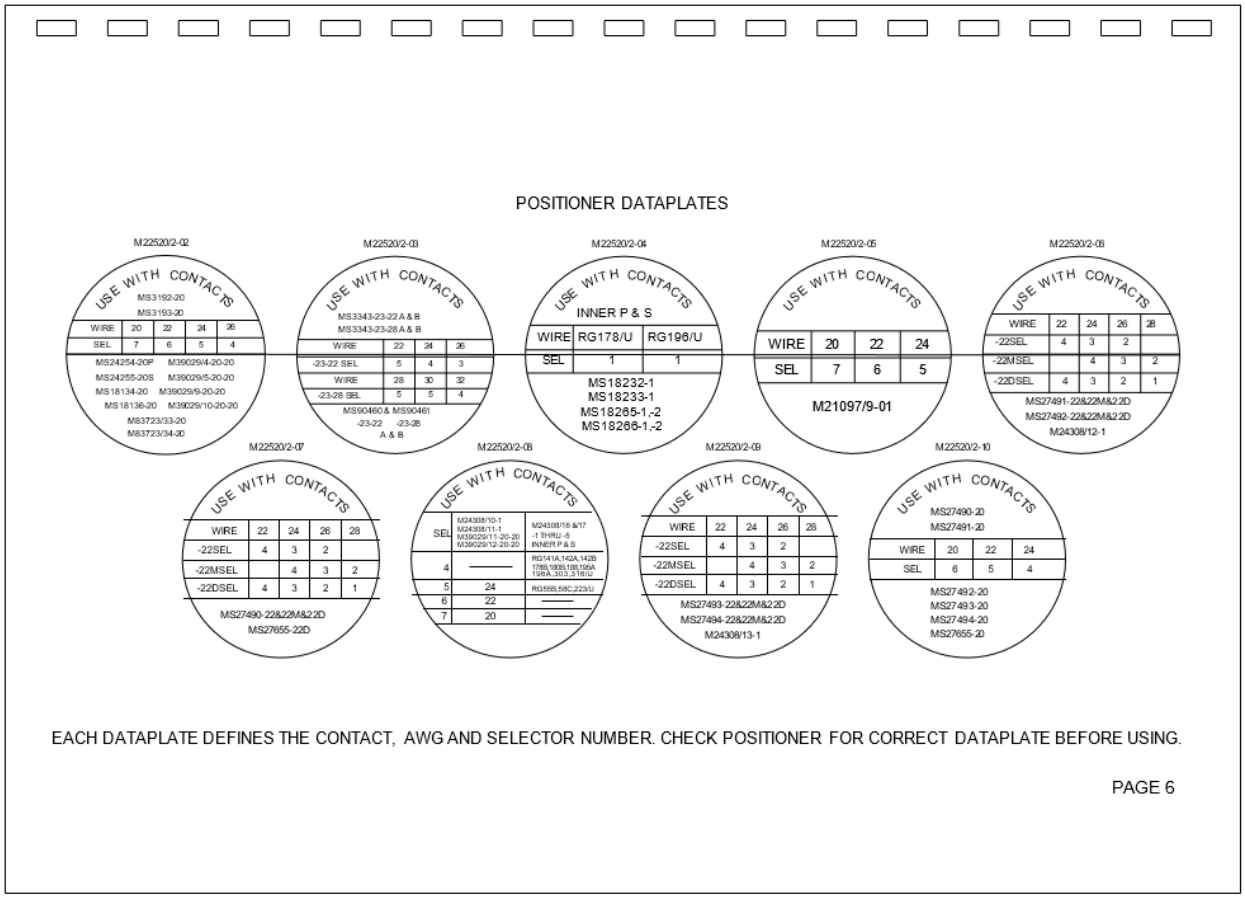
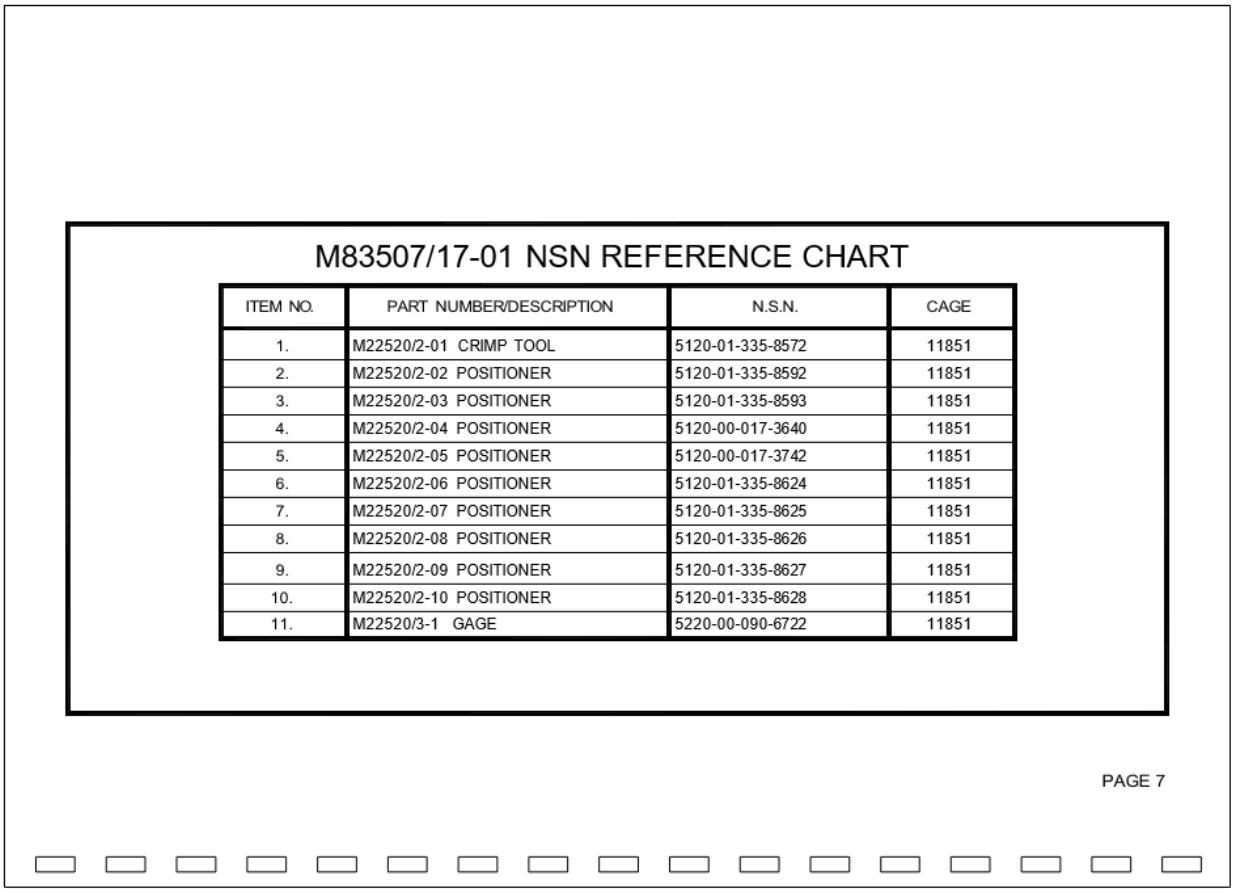


FIGURE 21 Plastic case. Instruction sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.



M83507/17-01 NSN REFERENCE CHART

ITEM NO.	PART NUMBER/DESCRIPTION	N.S.N.	CAGE
1.	M22520/2-01 CRIMP TOOL	5120-01-335-8572	11851
2.	M22520/2-02 POSITIONER	5120-01-335-8592	11851
3.	M22520/2-03 POSITIONER	5120-01-335-8593	11851
4.	M22520/2-04 POSITIONER	5120-00-017-3640	11851
5.	M22520/2-05 POSITIONER	5120-00-017-3742	11851
6.	M22520/2-06 POSITIONER	5120-01-335-8624	11851
7.	M22520/2-07 POSITIONER	5120-01-335-8625	11851
8.	M22520/2-08 POSITIONER	5120-01-335-8626	11851
9.	M22520/2-09 POSITIONER	5120-01-335-8627	11851
10.	M22520/2-10 POSITIONER	5120-01-335-8628	11851
11.	M22520/3-1 GAGE	5220-00-090-6722	11851

PAGE 7

FIGURE 22 Plastic case. Instruction sheet.

REQUIREMENTS:

Materials:

Case and cover (category 3):

Sheet metal, 21 gage in accordance with ASTM-A794/A794M; gray baked enamel on all metal parts.

Pelican plastic case

Instruction sheets and contents chart: White vinyl 0.020 inch (0.51 mm) thick. Insert and retainer pad: Fire retardant foamed polyethylene in accordance with A-A-59135 or A-A-59136; color is optional.

Clasps, draw pull type with provision for locking device, handles, and hinges: Suitably plated to resist corrosion.

Nameplate: In accordance with MIL-DTL-19834.

Binder: Looseleaf, three ring, 0.562 inch (14.27 mm) ±0.062 inch (1.57 mm) I.D., 2.75 inch (69.9 mm) center distance between rings, securely riveted to inside of cover.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.

Instruction sheets and contents chart (see pages 4 through 9): Lettering shall be legible and protected with a transparent coating. Contents chart (centered on inside of case cove) shall have pressure sensitive adhesive backing and show the insert cavity configurations and the respective tool part numbers; a list of NSN's will also be supplied.

Part number:

M83507/17-01 Tool Kit for SAE-AS22520/2 crimping tool and positioners.
M83507/17-02 Tool case assembly (tools not included).

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83507, this document references the following:

A-A-59135
A-A-59136
ASTM-A794/A794M
MIL-DTL-19834
SAE AS22520/2

CONCLUDING MATERIAL

Custodians:
Navy – AS

Preparing activity:
DLA - CC

Review activities:
Navy – EC, MC
DLA - GS

(Project 5180-2025-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:24Z
Check the source to verify that this is the current version before use.